

The "How-To" as a Post-AI Assessment Tool

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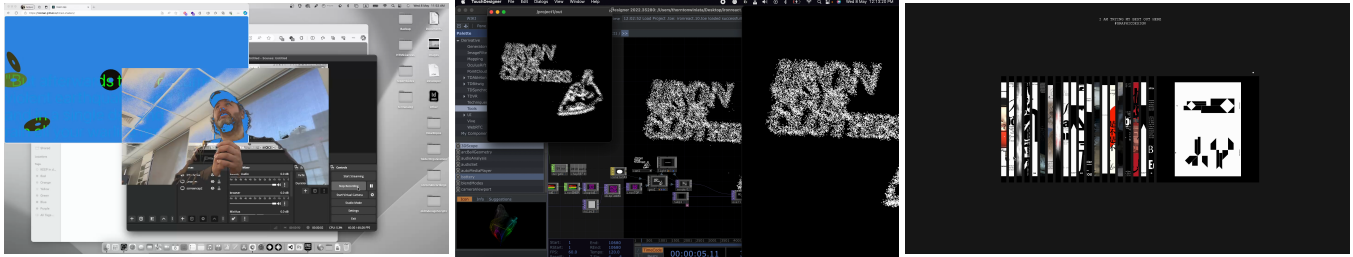


Figure 1: Example "How-Tos" from Whitecliffe College, School of Design Innovation, Interaction Design II, for the assignment: Experimental "How-To" Technology Guide, 2024. From left to right: Roman Mitch (example), Thornton Winiata, Claudia Yin.

ABSTRACT

This paper introduces a simple but innovative assignment technique, the "How-To" as a way to simplify the relationship with AI in assessment events. As an assignment, the "How-To" guide merges reflective practice with technical skill development in technical or creative fields, inviting students to author a guide based on insights from prior coursework, personal projects or specific directed challenges. This method emphasizes reciprocal learning by engaging students in a reflective process where they must articulate and analyze their technical learning, thus reinforcing their understanding and evidencing their mastery of the given learning outcome. This technique could be more broadly applicable, building on the large success of online how-to tutorials; but in flipping the emphasis from passive consumption to productive engagement, the discussed assignment encourages a blend of narrative, technical detail, and creative expression, culminating in a resource that benefits both the author, their peers and potentially course development itself. This presentation will explore how the assignment's framework develops as an extension of the principle of 'ako', a Māori concept emphasizing reciprocal learning. The discussion will focus on its application within the specific educational context of Aotearoa New Zealand and the broader value it offers to creative and technical education.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods; Blogs;** • **Social and professional topics** → **Computational thinking; Computing education programs.**

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KEYWORDS

Education, Undergraduate, Design, Interaction, Creative Coding, Critical Analysis, Multimedia

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1 OVERVIEW

Recent developments in artificial intelligence (AI) has ushered in unprecedented opportunities for learning and innovation, including at the intersection of creative coding and interactive techniques. Traditional educational assessments such as essays, exams, and reports often fall short in capturing the depth of student learning and creativity. At its worst this can mean that energy is spent on trying to ensure students are not cheating by using AI, where AI could be a perfect partner in the skills-acquisition process. This paper introduces an inventive assessment methodology that leverages "How-To" guides as reflective tools, encouraging students to critically engage with their learning experiences. By synthesizing insights from prior coursework, personal projects, or specific challenges, this approach not only reinforces technical competencies but also fosters a deeper understanding of creative processes. This initiative reflects a broader pedagogical shift towards active learning and knowledge co-creation, challenging students to articulate their learning journey in a manner that benefits the wider educational community.

Computational thinking is increasingly recognized as a foundational skill [Resnick et al. 2009]. The difficulty in teaching people the art of programming is well documented. For example introductory programming courses have a mean worldwide pass rate of 67.7% [Li and Watson 2014]. While cognitive abilities like spatial and logical reasoning are often highlighted [Jones and Jones

2008], factors such as class size and self-belief are more indicative of student success [Li and Watson 2014]. AI-assisted tools, such as GitHub Copilot, offer significant potential in demystifying the syntactical complexities of programming [GitHub 2021].

2 A NEW DOUBT IN THE ESSAY

Traditional educational assessments often fall short in capturing the depth of student learning and creativity. Essays for example, while valuable as a way to improve critical thinking, also introduce complexity by assessing more than just the learning outcome, including students' writing skills, critical thinking, and even their familiarity with the assessment format itself. This multifaceted assessment can obscure the true measure of a student's understanding and technical proficiency and computational thinking.

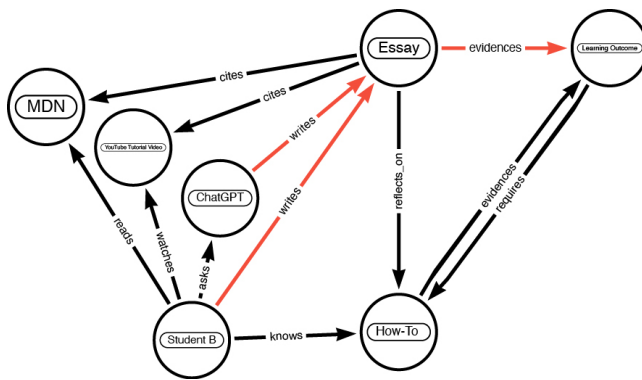


Figure 2: Graph representation of the post-AI dynamic of essays as assessment tools. Red edges represent relationships for which doubt enters the assessment context.

Moreover, with the rise of AI, the oft-cited concerns about academic integrity further complicate the essay's role as a fair and valid assessment tool. The capabilities of AI to simulate structured thinking could be said to introduce doubt into the network of the essay as an index of student learning [Abd-Elal and E. 2019]. As shown in Figure 2, the uncertainty of the student writing an essay also affects the essay as evidence of a given learning outcome.

3 ASSESSMENT THROUGH "HOW-TO" GUIDES

The "How-To" guide assignment offers a refreshing alternative to traditional essays, focusing on practical application and reflective practice. This method encourages students to articulate their learning process and technical skills through the creation of a step-by-step guide.

As shown in Figure 3, the "How-To" guide opens up a new set of possibilities and a simplified dynamic with AI and assessment.

"How-Tos":

- Require a willingness on the part of the teacher to "go along" with the students' learning.
- Promote the development of key skills such as pattern recognition, decomposition, and abstraction, crucial for computational thinking.

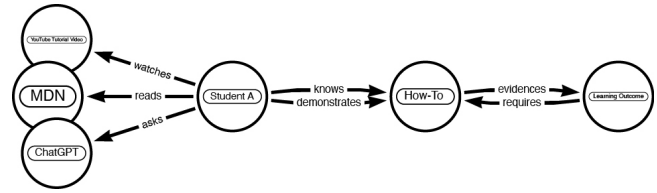


Figure 3: A graph representation of the simplified dynamic of "how-tos" as assessment tools.

- Reinforce technical competencies by requiring students to explain their processes, which aids in consolidating their understanding.
- Promote active learning and knowledge co-creation.
- Enhance engagement with AI tools as partners in the learning process [Acar 2023].

4 "HOW-TO" AS EXTENSION OF TE AO MĀORI

The Māori concept of "ako" emphasizes reciprocal learning and knowledge sharing. This approach values the student's voice and perspective in concert with the teacher. In this context knowledge is not perceived as a transactional deposit from an authority [Freire 2000], but a collaborative work inside a structured and highly principled relation [Pohatu 2004]. By allowing students to author "How-To" guides based on their unique insights and experiences, the assignment honors the principles of Te Ao Māori, promoting a holistic and inclusive educational framework.

5 CONCLUSION

The "How-To" guide method transcends traditional assessment methods like exams, which are limited in evaluating practical skills, and essays, which assess writing and argumentation skills but may not effectively measure technical proficiency. "How-To" guides require each student to articulate their learning and technical processes, offering a direct assessment of individual skills. This method fosters both collaborative and independent learning, supporting practical application and reflective practice—key in education in a post-AI world.

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